

# AQUATIC CONNECTIVITY Stream Crossing Survey DATA FORM

DATABASE ENTRY BY \_\_\_\_\_

ENTRY DATE \_\_\_\_\_

DATA ENTRY REVIEWED BY \_\_\_\_\_

REVIEW DATE \_\_\_\_\_

## CROSSING DATA

Crossing Code \_\_\_\_\_ Local ID (Optional) \_\_\_\_\_

Date Observed (00/00/0000) \_\_\_\_\_ Lead Observer \_\_\_\_\_

Town/County \_\_\_\_\_ Stream \_\_\_\_\_

Road \_\_\_\_\_ Type ☐ MULTILANE ☐ PAVED ☐ UNPAVED ☐ DRIVEWAY ☐ TRAIL ☐ RAILROADGPS Coordinates (Decimal degrees)        °N Latitude —        °W Longitude

### Location Description

|                                                                                                                                                                                                                                       |                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| Crossing Type <input type="checkbox"/> BRIDGE <input type="checkbox"/> CULVERT <input type="checkbox"/> MULTIPLE CULVERT <input type="checkbox"/> FORD <input type="checkbox"/> NO CROSSING <input type="checkbox"/> REMOVED CROSSING | Number of Culverts/ Bridge Cells<br>_____ |
| <input type="checkbox"/> BURIED STREAM <input type="checkbox"/> INACCESSIBLE <input type="checkbox"/> PARTIALLY INACCESSIBLE <input type="checkbox"/> NO UPSTREAM CHANNEL <input type="checkbox"/> BRIDGE ADEQUATE                    |                                           |

Photo IDs INLET \_\_\_\_\_ OUTLET \_\_\_\_\_ UPSTREAM \_\_\_\_\_ DOWNSTREAM \_\_\_\_\_ OTHER \_\_\_\_\_

Flow Condition ☐ NO FLOW ☐ TYPICAL-LOW ☐ MODERATE ☐ HIGH Crossing Condition ☐ OK ☐ POOR ☐ NEW ☐ UNKNOWNTidal Site ☐ YES ☐ NO ☐ UNKNOWN Alignment ☐ FLOW-ALIGNED ☐ SKEWED (>45°) Road Fill Height (Top of culvert to road surface; bridge = 0) \_\_\_\_\_Bankfull Width (Optional) \_\_\_\_\_ Confidence ☐ HIGH ☐ LOW/ESTIMATED Constriction ☐ SEVERE ☐ MODERATE ☐ SPANS ONLY BANKFULL/ACTIVE CHANNELTailwater Scour Pool ☐ NONE ☐ SMALL ☐ LARGE ☐ SPANS FULL CHANNEL & BANKS

### Crossing Comments

## STRUCTURE 1

Structure Material ☐ METAL ☐ CONCRETE ☐ PLASTIC ☐ WOOD ☐ ROCK/STONE ☐ FIBERGLASS ☐ COMBINATION

## OUTLET

Outlet Shape ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ FORD ☐ UNKNOWN ☐ REMOVED Outlet Armoring ☐ NONE ☐ NOT EXTENSIVE ☐ EXTENSIVEOutlet Grade (Pick one) ☐ AT STREAM GRADE ☐ FREE FALL ☐ CASCADE ☐ FREE FALL ONTO CASCADE ☐ CLOGGED/COLLAPSED/SUBMERGED ☐ UNKNOWN

Outlet Dimensions A. Width \_\_\_\_\_ B. Height \_\_\_\_\_ C. Substrate/Water Width \_\_\_\_\_ D. Water Depth \_\_\_\_\_

Outlet Drop to Water Surface \_\_\_\_\_ Outlet Drop to Stream Bottom \_\_\_\_\_ E. Abutment Height (Type 7 bridges only) \_\_\_\_\_

L. Structure Length (Overall length from inlet to outlet) \_\_\_\_\_

## INLET

Inlet Shape ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ FORD ☐ UNKNOWN ☐ REMOVEDInlet Type ☐ PROJECTING ☐ HEADWALL ☐ WINGWALLS ☐ HEADWALL & WINGWALLS ☐ MITERED TO SLOPE ☐ OTHER ☐ NONEInlet Grade (Pick one) ☐ AT STREAM GRADE ☐ INLET DROP ☐ PERCHED ☐ CLOGGED/COLLAPSED/SUBMERGED ☐ UNKNOWN

Inlet Dimensions A. Width \_\_\_\_\_ B. Height \_\_\_\_\_ C. Substrate/Water Width \_\_\_\_\_ D. Water Depth \_\_\_\_\_

## ADDITIONAL CONDITIONS

Slope % (Optional) \_\_\_\_\_ Slope Confidence ☐ HIGH ☐ LOW Internal Structures ☐ NONE ☐ BAFFLES/WEIRS ☐ SUPPORTS ☐ OTHER \_\_\_\_\_Structure Substrate Matches Stream ☐ NONE ☐ COMPARABLE ☐ CONTRASTING ☐ NOT APPROPRIATE ☐ UNKNOWNStructure Substrate Type (Pick one) ☐ NONE ☐ SILT ☐ SAND ☐ GRAVEL ☐ COBBLE ☐ BOULDER ☐ BEDROCK ☐ UNKNOWNStructure Substrate Coverage ☐ NONE ☐ 25% ☐ 50% ☐ 75% ☐ 100% ☐ UNKNOWNPhysical Barriers (Pick all that apply) ☐ NONE ☐ DEBRIS/SEDIMENT/ROCK ☐ DEFORMATION ☐ FREE FALL ☐ FENCING ☐ DRY ☐ OTHERSeverity (Choose carefully based on barrier type(s) above) ☐ NONE ☐ MINOR ☐ MODERATE ☐ SEVEREWater Depth Matches Stream ☐ YES ☐ NO-SHALLOWER ☐ NO-DEEPER ☐ UNKNOWN ☐ DRYWater Velocity Matches Stream ☐ YES ☐ NO-FASTER ☐ NO-SLOWER ☐ UNKNOWN ☐ DRYDry Passage through Structure? ☐ YES ☐ NO ☐ UNKNOWN Height above Dry Passage \_\_\_\_\_

### Comments

5/26/16

| STRUCTURE 2           |                                                            | Structure Material                                                                                                                                                                                                                                            |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|-----------------------|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|------------------------------|----------------------------|----------------------------------|------------------------------|------------------------------------------|-------------------------------|----------------------------------|----------------------------------|-----------------|-------------------------------|----------------------------------------|------------------------------------|-------------------------------|----------------------------------------|-----------------------------------|--------------------------------|
| OUTLET                | Outlet Shape                                               | <input type="checkbox"/> 1                                                                                                                                                                                                                                    | <input type="checkbox"/> 2 | <input type="checkbox"/> 3   | <input type="checkbox"/> 4 | <input type="checkbox"/> 5       | <input type="checkbox"/> 6   | <input type="checkbox"/> 7               | <input type="checkbox"/> FORD | <input type="checkbox"/> UNKNOWN | <input type="checkbox"/> REMOVED | Outlet Armoring | <input type="checkbox"/> NONE | <input type="checkbox"/> NOT EXTENSIVE | <input type="checkbox"/> EXTENSIVE |                               |                                        |                                   |                                |
|                       | Outlet Grade (Pick one)                                    | <input type="checkbox"/> AT STREAM GRADE <input type="checkbox"/> FREE FALL <input type="checkbox"/> CASCADE <input type="checkbox"/> FREE FALL ONTO CASCADE <input type="checkbox"/> CLOGGED/COLLAPSED/SUBMERGED <input type="checkbox"/> UNKNOWN            |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Outlet Dimensions                                          | A. Width                                                                                                                                                                                                                                                      |                            | B. Height                    |                            | C. Substrate/Water Width         |                              |                                          |                               | D. Water Depth                   |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Outlet Drop to Water Surface                               |                                                                                                                                                                                                                                                               |                            | Outlet Drop to Stream Bottom |                            |                                  |                              | E. Abutment Height (Type 7 bridges only) |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | L. Structure Length (Overall length from inlet to outlet)  |                                                                                                                                                                                                                                                               |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
| INLET                 | Inlet Shape                                                | <input type="checkbox"/> 1                                                                                                                                                                                                                                    | <input type="checkbox"/> 2 | <input type="checkbox"/> 3   | <input type="checkbox"/> 4 | <input type="checkbox"/> 5       | <input type="checkbox"/> 6   | <input type="checkbox"/> 7               | <input type="checkbox"/> FORD | <input type="checkbox"/> UNKNOWN | <input type="checkbox"/> REMOVED |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Inlet Type                                                 | <input type="checkbox"/> PROJECTING <input type="checkbox"/> HEADWALL <input type="checkbox"/> WINGWALLS <input type="checkbox"/> HEADWALL & WINGWALLS <input type="checkbox"/> MITERED TO SLOPE <input type="checkbox"/> OTHER <input type="checkbox"/> NONE |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Inlet Grade (Pick one)                                     | <input type="checkbox"/> AT STREAM GRADE <input type="checkbox"/> INLET DROP <input type="checkbox"/> PERCHED <input type="checkbox"/> CLOGGED/COLLAPSED/SUBMERGED <input type="checkbox"/> UNKNOWN                                                           |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Inlet Dimensions                                           | A. Width                                                                                                                                                                                                                                                      |                            | B. Height                    |                            | C. Substrate/Water Width         |                              |                                          |                               | D. Water Depth                   |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
| ADDITIONAL CONDITIONS | Slope % (Optional)                                         |                                                                                                                                                                                                                                                               |                            | Slope Confidence             |                            | <input type="checkbox"/> HIGH    | <input type="checkbox"/> LOW | Internal Structures                      |                               |                                  |                                  |                 |                               |                                        |                                    | <input type="checkbox"/> NONE | <input type="checkbox"/> BAFFLES/WEIRS | <input type="checkbox"/> SUPPORTS | <input type="checkbox"/> OTHER |
|                       | Structure Substrate Matches Stream                         | <input type="checkbox"/> NONE <input type="checkbox"/> COMPARABLE <input type="checkbox"/> CONTRASTING <input type="checkbox"/> NOT APPROPRIATE <input type="checkbox"/> UNKNOWN                                                                              |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Structure Substrate Type (Pick one)                        | <input type="checkbox"/> NONE <input type="checkbox"/> SILT <input type="checkbox"/> SAND <input type="checkbox"/> GRAVEL <input type="checkbox"/> COBBLE <input type="checkbox"/> BOULDER <input type="checkbox"/> BEDROCK <input type="checkbox"/> UNKNOWN  |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Structure Substrate Coverage                               | <input type="checkbox"/> NONE <input type="checkbox"/> 25% <input type="checkbox"/> 50% <input type="checkbox"/> 75% <input type="checkbox"/> 100% <input type="checkbox"/> UNKNOWN                                                                           |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Physical Barriers (Pick all that apply)                    | <input type="checkbox"/> NONE <input type="checkbox"/> DEBRIS/SEDIMENT/ROCK <input type="checkbox"/> DEFORMATION <input type="checkbox"/> FREE FALL <input type="checkbox"/> FENCING <input type="checkbox"/> DRY <input type="checkbox"/> OTHER              |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Severity (Choose carefully based on barrier type(s) above) | <input type="checkbox"/> NONE <input type="checkbox"/> MINOR <input type="checkbox"/> MODERATE <input type="checkbox"/> SEVERE                                                                                                                                |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Water Depth Matches Stream                                 | <input type="checkbox"/> YES <input type="checkbox"/> NO-SHALLOWER <input type="checkbox"/> NO-DEEPER <input type="checkbox"/> UNKNOWN <input type="checkbox"/> DRY                                                                                           |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Water Velocity Matches Stream                              | <input type="checkbox"/> YES <input type="checkbox"/> NO-FASTER <input type="checkbox"/> NO-SLOWER <input type="checkbox"/> UNKNOWN <input type="checkbox"/> DRY                                                                                              |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Dry Passage through Structure?                             | <input type="checkbox"/> YES                                                                                                                                                                                                                                  |                            | <input type="checkbox"/> NO  |                            | <input type="checkbox"/> UNKNOWN |                              | Height above Dry Passage                 |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Comments                                                   |                                                                                                                                                                                                                                                               |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |

| STRUCTURE 3           |                                                            | Structure Material                                                                                                                                                                                                                                            |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|-----------------------|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|------------------------------|----------------------------|----------------------------------|------------------------------|------------------------------------------|-------------------------------|----------------------------------|----------------------------------|-----------------|-------------------------------|----------------------------------------|------------------------------------|-------------------------------|----------------------------------------|-----------------------------------|--------------------------------|
| OUTLET                | Outlet Shape                                               | <input type="checkbox"/> 1                                                                                                                                                                                                                                    | <input type="checkbox"/> 2 | <input type="checkbox"/> 3   | <input type="checkbox"/> 4 | <input type="checkbox"/> 5       | <input type="checkbox"/> 6   | <input type="checkbox"/> 7               | <input type="checkbox"/> FORD | <input type="checkbox"/> UNKNOWN | <input type="checkbox"/> REMOVED | Outlet Armoring | <input type="checkbox"/> NONE | <input type="checkbox"/> NOT EXTENSIVE | <input type="checkbox"/> EXTENSIVE |                               |                                        |                                   |                                |
|                       | Outlet Grade (Pick one)                                    | <input type="checkbox"/> AT STREAM GRADE <input type="checkbox"/> FREE FALL <input type="checkbox"/> CASCADE <input type="checkbox"/> FREE FALL ONTO CASCADE <input type="checkbox"/> CLOGGED/COLLAPSED/SUBMERGED <input type="checkbox"/> UNKNOWN            |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Outlet Dimensions                                          | A. Width                                                                                                                                                                                                                                                      |                            | B. Height                    |                            | C. Substrate/Water Width         |                              |                                          |                               | D. Water Depth                   |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Outlet Drop to Water Surface                               |                                                                                                                                                                                                                                                               |                            | Outlet Drop to Stream Bottom |                            |                                  |                              | E. Abutment Height (Type 7 bridges only) |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | L. Structure Length (Overall length from inlet to outlet)  |                                                                                                                                                                                                                                                               |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
| INLET                 | Inlet Shape                                                | <input type="checkbox"/> 1                                                                                                                                                                                                                                    | <input type="checkbox"/> 2 | <input type="checkbox"/> 3   | <input type="checkbox"/> 4 | <input type="checkbox"/> 5       | <input type="checkbox"/> 6   | <input type="checkbox"/> 7               | <input type="checkbox"/> FORD | <input type="checkbox"/> UNKNOWN | <input type="checkbox"/> REMOVED |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Inlet Type                                                 | <input type="checkbox"/> PROJECTING <input type="checkbox"/> HEADWALL <input type="checkbox"/> WINGWALLS <input type="checkbox"/> HEADWALL & WINGWALLS <input type="checkbox"/> MITERED TO SLOPE <input type="checkbox"/> OTHER <input type="checkbox"/> NONE |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Inlet Grade (Pick one)                                     | <input type="checkbox"/> AT STREAM GRADE <input type="checkbox"/> INLET DROP <input type="checkbox"/> PERCHED <input type="checkbox"/> CLOGGED/COLLAPSED/SUBMERGED <input type="checkbox"/> UNKNOWN                                                           |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Inlet Dimensions                                           | A. Width                                                                                                                                                                                                                                                      |                            | B. Height                    |                            | C. Substrate/Water Width         |                              |                                          |                               | D. Water Depth                   |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
| ADDITIONAL CONDITIONS | Slope % (Optional)                                         |                                                                                                                                                                                                                                                               |                            | Slope Confidence             |                            | <input type="checkbox"/> HIGH    | <input type="checkbox"/> LOW | Internal Structures                      |                               |                                  |                                  |                 |                               |                                        |                                    | <input type="checkbox"/> NONE | <input type="checkbox"/> BAFFLES/WEIRS | <input type="checkbox"/> SUPPORTS | <input type="checkbox"/> OTHER |
|                       | Structure Substrate Matches Stream                         | <input type="checkbox"/> NONE <input type="checkbox"/> COMPARABLE <input type="checkbox"/> CONTRASTING <input type="checkbox"/> NOT APPROPRIATE <input type="checkbox"/> UNKNOWN                                                                              |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Structure Substrate Type (Pick one)                        | <input type="checkbox"/> NONE <input type="checkbox"/> SILT <input type="checkbox"/> SAND <input type="checkbox"/> GRAVEL <input type="checkbox"/> COBBLE <input type="checkbox"/> BOULDER <input type="checkbox"/> BEDROCK <input type="checkbox"/> UNKNOWN  |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Structure Substrate Coverage                               | <input type="checkbox"/> NONE <input type="checkbox"/> 25% <input type="checkbox"/> 50% <input type="checkbox"/> 75% <input type="checkbox"/> 100% <input type="checkbox"/> UNKNOWN                                                                           |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Physical Barriers (Pick all that apply)                    | <input type="checkbox"/> NONE <input type="checkbox"/> DEBRIS/SEDIMENT/ROCK <input type="checkbox"/> DEFORMATION <input type="checkbox"/> FREE FALL <input type="checkbox"/> FENCING <input type="checkbox"/> DRY <input type="checkbox"/> OTHER              |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Severity (Choose carefully based on barrier type(s) above) | <input type="checkbox"/> NONE <input type="checkbox"/> MINOR <input type="checkbox"/> MODERATE <input type="checkbox"/> SEVERE                                                                                                                                |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Water Depth Matches Stream                                 | <input type="checkbox"/> YES <input type="checkbox"/> NO-SHALLOWER <input type="checkbox"/> NO-DEEPER <input type="checkbox"/> UNKNOWN <input type="checkbox"/> DRY                                                                                           |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Water Velocity Matches Stream                              | <input type="checkbox"/> YES <input type="checkbox"/> NO-FASTER <input type="checkbox"/> NO-SLOWER <input type="checkbox"/> UNKNOWN <input type="checkbox"/> DRY                                                                                              |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Dry Passage through Structure?                             | <input type="checkbox"/> YES                                                                                                                                                                                                                                  |                            | <input type="checkbox"/> NO  |                            | <input type="checkbox"/> UNKNOWN |                              | Height above Dry Passage                 |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Comments                                                   |                                                                                                                                                                                                                                                               |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |

| STRUCTURE 4           |                                                            | Structure Material                                                                                                                                                                                                                                            |                            |                                    |                            |                                  |                              |                                                |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                     |
|-----------------------|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|------------------------------------|----------------------------|----------------------------------|------------------------------|------------------------------------------------|-------------------------------|----------------------------------|----------------------------------|-----------------|-------------------------------|----------------------------------------|------------------------------------|-------------------------------|----------------------------------------|-----------------------------------|-------------------------------------|
| OUTLET                | Outlet Shape                                               | <input type="checkbox"/> 1                                                                                                                                                                                                                                    | <input type="checkbox"/> 2 | <input type="checkbox"/> 3         | <input type="checkbox"/> 4 | <input type="checkbox"/> 5       | <input type="checkbox"/> 6   | <input type="checkbox"/> 7                     | <input type="checkbox"/> FORD | <input type="checkbox"/> UNKNOWN | <input type="checkbox"/> REMOVED | Outlet Armoring | <input type="checkbox"/> NONE | <input type="checkbox"/> NOT EXTENSIVE | <input type="checkbox"/> EXTENSIVE |                               |                                        |                                   |                                     |
|                       | Outlet Grade (Pick one)                                    | <input type="checkbox"/> AT STREAM GRADE <input type="checkbox"/> FREE FALL <input type="checkbox"/> CASCADE <input type="checkbox"/> FREE FALL ONTO CASCADE <input type="checkbox"/> CLOGGED/COLLAPSED/SUBMERGED <input type="checkbox"/> UNKNOWN            |                            |                                    |                            |                                  |                              |                                                |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                     |
|                       | Outlet Dimensions                                          | A. Width_____.                                                                                                                                                                                                                                                |                            | B. Height_____.                    |                            | C. Substrate/Water Width_____.   |                              |                                                |                               | D. Water Depth_____.             |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                     |
|                       | Outlet Drop to Water Surface                               | _____.                                                                                                                                                                                                                                                        |                            | Outlet Drop to Stream Bottom_____. |                            |                                  |                              | E. Abutment Height (Type 7 bridges only)_____. |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                     |
|                       | L. Structure Length (Overall length from inlet to outlet)  | _____.                                                                                                                                                                                                                                                        |                            |                                    |                            |                                  |                              |                                                |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                     |
| INLET                 | Inlet Shape                                                | <input type="checkbox"/> 1                                                                                                                                                                                                                                    | <input type="checkbox"/> 2 | <input type="checkbox"/> 3         | <input type="checkbox"/> 4 | <input type="checkbox"/> 5       | <input type="checkbox"/> 6   | <input type="checkbox"/> 7                     | <input type="checkbox"/> FORD | <input type="checkbox"/> UNKNOWN | <input type="checkbox"/> REMOVED |                 |                               |                                        |                                    |                               |                                        |                                   |                                     |
|                       | Inlet Type                                                 | <input type="checkbox"/> PROJECTING <input type="checkbox"/> HEADWALL <input type="checkbox"/> WINGWALLS <input type="checkbox"/> HEADWALL & WINGWALLS <input type="checkbox"/> MITERED TO SLOPE <input type="checkbox"/> OTHER <input type="checkbox"/> NONE |                            |                                    |                            |                                  |                              |                                                |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                     |
|                       | Inlet Grade (Pick one)                                     | <input type="checkbox"/> AT STREAM GRADE <input type="checkbox"/> INLET DROP <input type="checkbox"/> PERCHED <input type="checkbox"/> CLOGGED/COLLAPSED/SUBMERGED <input type="checkbox"/> UNKNOWN                                                           |                            |                                    |                            |                                  |                              |                                                |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                     |
|                       | Inlet Dimensions                                           | A. Width_____.                                                                                                                                                                                                                                                |                            | B. Height_____.                    |                            | C. Substrate/Water Width_____.   |                              |                                                |                               | D. Water Depth_____.             |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                     |
| ADDITIONAL CONDITIONS | Slope % (Optional)                                         | _____                                                                                                                                                                                                                                                         |                            | Slope Confidence                   |                            | <input type="checkbox"/> HIGH    | <input type="checkbox"/> LOW | Internal Structures                            |                               |                                  |                                  |                 |                               |                                        |                                    | <input type="checkbox"/> NONE | <input type="checkbox"/> BAFFLES/WEIRS | <input type="checkbox"/> SUPPORTS | <input type="checkbox"/> OTHER_____ |
|                       | Structure Substrate Matches Stream                         | <input type="checkbox"/> NONE <input type="checkbox"/> COMPARABLE <input type="checkbox"/> CONTRASTING <input type="checkbox"/> NOT APPROPRIATE <input type="checkbox"/> UNKNOWN                                                                              |                            |                                    |                            |                                  |                              |                                                |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                     |
|                       | Structure Substrate Type (Pick one)                        | <input type="checkbox"/> NONE <input type="checkbox"/> SILT <input type="checkbox"/> SAND <input type="checkbox"/> GRAVEL <input type="checkbox"/> COBBLE <input type="checkbox"/> BOULDER <input type="checkbox"/> BEDROCK <input type="checkbox"/> UNKNOWN  |                            |                                    |                            |                                  |                              |                                                |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                     |
|                       | Structure Substrate Coverage                               | <input type="checkbox"/> NONE <input type="checkbox"/> 25% <input type="checkbox"/> 50% <input type="checkbox"/> 75% <input type="checkbox"/> 100% <input type="checkbox"/> UNKNOWN                                                                           |                            |                                    |                            |                                  |                              |                                                |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                     |
|                       | Physical Barriers (Pick all that apply)                    | <input type="checkbox"/> NONE <input type="checkbox"/> DEBRIS/SEDIMENT/ROCK <input type="checkbox"/> DEFORMATION <input type="checkbox"/> FREE FALL <input type="checkbox"/> FENCING <input type="checkbox"/> DRY <input type="checkbox"/> OTHER              |                            |                                    |                            |                                  |                              |                                                |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                     |
|                       | Severity (Choose carefully based on barrier type(s) above) | <input type="checkbox"/> NONE <input type="checkbox"/> MINOR <input type="checkbox"/> MODERATE <input type="checkbox"/> SEVERE                                                                                                                                |                            |                                    |                            |                                  |                              |                                                |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                     |
|                       | Water Depth Matches Stream                                 | <input type="checkbox"/> YES <input type="checkbox"/> NO-SHALLOWER <input type="checkbox"/> NO-DEEPER <input type="checkbox"/> UNKNOWN <input type="checkbox"/> DRY                                                                                           |                            |                                    |                            |                                  |                              |                                                |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                     |
|                       | Water Velocity Matches Stream                              | <input type="checkbox"/> YES <input type="checkbox"/> NO-FASTER <input type="checkbox"/> NO-SLOWER <input type="checkbox"/> UNKNOWN <input type="checkbox"/> DRY                                                                                              |                            |                                    |                            |                                  |                              |                                                |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                     |
|                       | Dry Passage through Structure?                             | <input type="checkbox"/> YES                                                                                                                                                                                                                                  |                            | <input type="checkbox"/> NO        |                            | <input type="checkbox"/> UNKNOWN |                              | Height above Dry Passage_____                  |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                     |
|                       | Comments                                                   |                                                                                                                                                                                                                                                               |                            |                                    |                            |                                  |                              |                                                |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                     |

| STRUCTURE 5           |                                                            | Structure Material                                                                                                                                                                                                                                            |                            |                                    |                            |                                  |                              |                                                |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                     |
|-----------------------|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|------------------------------------|----------------------------|----------------------------------|------------------------------|------------------------------------------------|-------------------------------|----------------------------------|----------------------------------|-----------------|-------------------------------|----------------------------------------|------------------------------------|-------------------------------|----------------------------------------|-----------------------------------|-------------------------------------|
| OUTLET                | Outlet Shape                                               | <input type="checkbox"/> 1                                                                                                                                                                                                                                    | <input type="checkbox"/> 2 | <input type="checkbox"/> 3         | <input type="checkbox"/> 4 | <input type="checkbox"/> 5       | <input type="checkbox"/> 6   | <input type="checkbox"/> 7                     | <input type="checkbox"/> FORD | <input type="checkbox"/> UNKNOWN | <input type="checkbox"/> REMOVED | Outlet Armoring | <input type="checkbox"/> NONE | <input type="checkbox"/> NOT EXTENSIVE | <input type="checkbox"/> EXTENSIVE |                               |                                        |                                   |                                     |
|                       | Outlet Grade (Pick one)                                    | <input type="checkbox"/> AT STREAM GRADE <input type="checkbox"/> FREE FALL <input type="checkbox"/> CASCADE <input type="checkbox"/> FREE FALL ONTO CASCADE <input type="checkbox"/> CLOGGED/COLLAPSED/SUBMERGED <input type="checkbox"/> UNKNOWN            |                            |                                    |                            |                                  |                              |                                                |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                     |
|                       | Outlet Dimensions                                          | A. Width_____.                                                                                                                                                                                                                                                |                            | B. Height_____.                    |                            | C. Substrate/Water Width_____.   |                              |                                                |                               | D. Water Depth_____.             |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                     |
|                       | Outlet Drop to Water Surface                               | _____.                                                                                                                                                                                                                                                        |                            | Outlet Drop to Stream Bottom_____. |                            |                                  |                              | E. Abutment Height (Type 7 bridges only)_____. |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                     |
|                       | L. Structure Length (Overall length from inlet to outlet)  | _____.                                                                                                                                                                                                                                                        |                            |                                    |                            |                                  |                              |                                                |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                     |
| INLET                 | Inlet Shape                                                | <input type="checkbox"/> 1                                                                                                                                                                                                                                    | <input type="checkbox"/> 2 | <input type="checkbox"/> 3         | <input type="checkbox"/> 4 | <input type="checkbox"/> 5       | <input type="checkbox"/> 6   | <input type="checkbox"/> 7                     | <input type="checkbox"/> FORD | <input type="checkbox"/> UNKNOWN | <input type="checkbox"/> REMOVED |                 |                               |                                        |                                    |                               |                                        |                                   |                                     |
|                       | Inlet Type                                                 | <input type="checkbox"/> PROJECTING <input type="checkbox"/> HEADWALL <input type="checkbox"/> WINGWALLS <input type="checkbox"/> HEADWALL & WINGWALLS <input type="checkbox"/> MITERED TO SLOPE <input type="checkbox"/> OTHER <input type="checkbox"/> NONE |                            |                                    |                            |                                  |                              |                                                |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                     |
|                       | Inlet Grade (Pick one)                                     | <input type="checkbox"/> AT STREAM GRADE <input type="checkbox"/> INLET DROP <input type="checkbox"/> PERCHED <input type="checkbox"/> CLOGGED/COLLAPSED/SUBMERGED <input type="checkbox"/> UNKNOWN                                                           |                            |                                    |                            |                                  |                              |                                                |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                     |
|                       | Inlet Dimensions                                           | A. Width_____.                                                                                                                                                                                                                                                |                            | B. Height_____.                    |                            | C. Substrate/Water Width_____.   |                              |                                                |                               | D. Water Depth_____.             |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                     |
| ADDITIONAL CONDITIONS | Slope % (Optional)                                         | _____                                                                                                                                                                                                                                                         |                            | Slope Confidence                   |                            | <input type="checkbox"/> HIGH    | <input type="checkbox"/> LOW | Internal Structures                            |                               |                                  |                                  |                 |                               |                                        |                                    | <input type="checkbox"/> NONE | <input type="checkbox"/> BAFFLES/WEIRS | <input type="checkbox"/> SUPPORTS | <input type="checkbox"/> OTHER_____ |
|                       | Structure Substrate Matches Stream                         | <input type="checkbox"/> NONE <input type="checkbox"/> COMPARABLE <input type="checkbox"/> CONTRASTING <input type="checkbox"/> NOT APPROPRIATE <input type="checkbox"/> UNKNOWN                                                                              |                            |                                    |                            |                                  |                              |                                                |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                     |
|                       | Structure Substrate Type (Pick one)                        | <input type="checkbox"/> NONE <input type="checkbox"/> SILT <input type="checkbox"/> SAND <input type="checkbox"/> GRAVEL <input type="checkbox"/> COBBLE <input type="checkbox"/> BOULDER <input type="checkbox"/> BEDROCK <input type="checkbox"/> UNKNOWN  |                            |                                    |                            |                                  |                              |                                                |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                     |
|                       | Structure Substrate Coverage                               | <input type="checkbox"/> NONE <input type="checkbox"/> 25% <input type="checkbox"/> 50% <input type="checkbox"/> 75% <input type="checkbox"/> 100% <input type="checkbox"/> UNKNOWN                                                                           |                            |                                    |                            |                                  |                              |                                                |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                     |
|                       | Physical Barriers (Pick all that apply)                    | <input type="checkbox"/> NONE <input type="checkbox"/> DEBRIS/SEDIMENT/ROCK <input type="checkbox"/> DEFORMATION <input type="checkbox"/> FREE FALL <input type="checkbox"/> FENCING <input type="checkbox"/> DRY <input type="checkbox"/> OTHER              |                            |                                    |                            |                                  |                              |                                                |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                     |
|                       | Severity (Choose carefully based on barrier type(s) above) | <input type="checkbox"/> NONE <input type="checkbox"/> MINOR <input type="checkbox"/> MODERATE <input type="checkbox"/> SEVERE                                                                                                                                |                            |                                    |                            |                                  |                              |                                                |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                     |
|                       | Water Depth Matches Stream                                 | <input type="checkbox"/> YES <input type="checkbox"/> NO-SHALLOWER <input type="checkbox"/> NO-DEEPER <input type="checkbox"/> UNKNOWN <input type="checkbox"/> DRY                                                                                           |                            |                                    |                            |                                  |                              |                                                |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                     |
|                       | Water Velocity Matches Stream                              | <input type="checkbox"/> YES <input type="checkbox"/> NO-FASTER <input type="checkbox"/> NO-SLOWER <input type="checkbox"/> UNKNOWN <input type="checkbox"/> DRY                                                                                              |                            |                                    |                            |                                  |                              |                                                |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                     |
|                       | Dry Passage through Structure?                             | <input type="checkbox"/> YES                                                                                                                                                                                                                                  |                            | <input type="checkbox"/> NO        |                            | <input type="checkbox"/> UNKNOWN |                              | Height above Dry Passage_____                  |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                     |
|                       | Comments                                                   |                                                                                                                                                                                                                                                               |                            |                                    |                            |                                  |                              |                                                |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                     |

| STRUCTURE 6           |                                                            | Structure Material                                                                                                                                                                                                                                            |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|-----------------------|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|------------------------------|----------------------------|----------------------------------|------------------------------|------------------------------------------|-------------------------------|----------------------------------|----------------------------------|-----------------|-------------------------------|----------------------------------------|------------------------------------|-------------------------------|----------------------------------------|-----------------------------------|--------------------------------|
| OUTLET                | Outlet Shape                                               | <input type="checkbox"/> 1                                                                                                                                                                                                                                    | <input type="checkbox"/> 2 | <input type="checkbox"/> 3   | <input type="checkbox"/> 4 | <input type="checkbox"/> 5       | <input type="checkbox"/> 6   | <input type="checkbox"/> 7               | <input type="checkbox"/> FORD | <input type="checkbox"/> UNKNOWN | <input type="checkbox"/> REMOVED | Outlet Armoring | <input type="checkbox"/> NONE | <input type="checkbox"/> NOT EXTENSIVE | <input type="checkbox"/> EXTENSIVE |                               |                                        |                                   |                                |
|                       | Outlet Grade (Pick one)                                    | <input type="checkbox"/> AT STREAM GRADE <input type="checkbox"/> FREE FALL <input type="checkbox"/> CASCADE <input type="checkbox"/> FREE FALL ONTO CASCADE <input type="checkbox"/> CLOGGED/COLLAPSED/SUBMERGED <input type="checkbox"/> UNKNOWN            |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Outlet Dimensions                                          | A. Width                                                                                                                                                                                                                                                      |                            | B. Height                    |                            | C. Substrate/Water Width         |                              |                                          | D. Water Depth                |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Outlet Drop to Water Surface                               |                                                                                                                                                                                                                                                               |                            | Outlet Drop to Stream Bottom |                            |                                  |                              | E. Abutment Height (Type 7 bridges only) |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | L. Structure Length (Overall length from inlet to outlet)  |                                                                                                                                                                                                                                                               |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
| INLET                 | Inlet Shape                                                | <input type="checkbox"/> 1                                                                                                                                                                                                                                    | <input type="checkbox"/> 2 | <input type="checkbox"/> 3   | <input type="checkbox"/> 4 | <input type="checkbox"/> 5       | <input type="checkbox"/> 6   | <input type="checkbox"/> 7               | <input type="checkbox"/> FORD | <input type="checkbox"/> UNKNOWN | <input type="checkbox"/> REMOVED |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Inlet Type                                                 | <input type="checkbox"/> PROJECTING <input type="checkbox"/> HEADWALL <input type="checkbox"/> WINGWALLS <input type="checkbox"/> HEADWALL & WINGWALLS <input type="checkbox"/> MITERED TO SLOPE <input type="checkbox"/> OTHER <input type="checkbox"/> NONE |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Inlet Grade (Pick one)                                     | <input type="checkbox"/> AT STREAM GRADE <input type="checkbox"/> INLET DROP <input type="checkbox"/> PERCHED <input type="checkbox"/> CLOGGED/COLLAPSED/SUBMERGED <input type="checkbox"/> UNKNOWN                                                           |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Inlet Dimensions                                           | A. Width                                                                                                                                                                                                                                                      |                            | B. Height                    |                            | C. Substrate/Water Width         |                              |                                          | D. Water Depth                |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
| ADDITIONAL CONDITIONS | Slope % (Optional)                                         |                                                                                                                                                                                                                                                               |                            | Slope Confidence             |                            | <input type="checkbox"/> HIGH    | <input type="checkbox"/> LOW | Internal Structures                      |                               |                                  |                                  |                 |                               |                                        |                                    | <input type="checkbox"/> NONE | <input type="checkbox"/> BAFFLES/WEIRS | <input type="checkbox"/> SUPPORTS | <input type="checkbox"/> OTHER |
|                       | Structure Substrate Matches Stream                         | <input type="checkbox"/> NONE <input type="checkbox"/> COMPARABLE <input type="checkbox"/> CONTRASTING <input type="checkbox"/> NOT APPROPRIATE <input type="checkbox"/> UNKNOWN                                                                              |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Structure Substrate Type (Pick one)                        | <input type="checkbox"/> NONE <input type="checkbox"/> SILT <input type="checkbox"/> SAND <input type="checkbox"/> GRAVEL <input type="checkbox"/> COBBLE <input type="checkbox"/> BOULDER <input type="checkbox"/> BEDROCK <input type="checkbox"/> UNKNOWN  |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Structure Substrate Coverage                               | <input type="checkbox"/> NONE <input type="checkbox"/> 25% <input type="checkbox"/> 50% <input type="checkbox"/> 75% <input type="checkbox"/> 100% <input type="checkbox"/> UNKNOWN                                                                           |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Physical Barriers (Pick all that apply)                    | <input type="checkbox"/> NONE <input type="checkbox"/> DEBRIS/SEDIMENT/ROCK <input type="checkbox"/> DEFORMATION <input type="checkbox"/> FREE FALL <input type="checkbox"/> FENCING <input type="checkbox"/> DRY <input type="checkbox"/> OTHER              |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Severity (Choose carefully based on barrier type(s) above) | <input type="checkbox"/> NONE <input type="checkbox"/> MINOR <input type="checkbox"/> MODERATE <input type="checkbox"/> SEVERE                                                                                                                                |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Water Depth Matches Stream                                 | <input type="checkbox"/> YES <input type="checkbox"/> NO-SHALLOWER <input type="checkbox"/> NO-DEEPER <input type="checkbox"/> UNKNOWN <input type="checkbox"/> DRY                                                                                           |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Water Velocity Matches Stream                              | <input type="checkbox"/> YES <input type="checkbox"/> NO-FASTER <input type="checkbox"/> NO-SLOWER <input type="checkbox"/> UNKNOWN <input type="checkbox"/> DRY                                                                                              |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Dry Passage through Structure?                             | <input type="checkbox"/> YES                                                                                                                                                                                                                                  |                            | <input type="checkbox"/> NO  |                            | <input type="checkbox"/> UNKNOWN |                              | Height above Dry Passage                 |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Comments                                                   |                                                                                                                                                                                                                                                               |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |

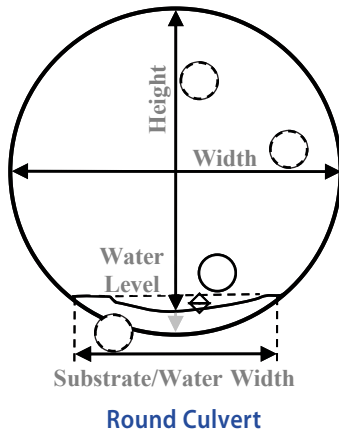
| STRUCTURE 7           |                                                            | Structure Material                                                                                                                                                                                                                                            |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|-----------------------|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|------------------------------|----------------------------|----------------------------------|------------------------------|------------------------------------------|-------------------------------|----------------------------------|----------------------------------|-----------------|-------------------------------|----------------------------------------|------------------------------------|-------------------------------|----------------------------------------|-----------------------------------|--------------------------------|
| OUTLET                | Outlet Shape                                               | <input type="checkbox"/> 1                                                                                                                                                                                                                                    | <input type="checkbox"/> 2 | <input type="checkbox"/> 3   | <input type="checkbox"/> 4 | <input type="checkbox"/> 5       | <input type="checkbox"/> 6   | <input type="checkbox"/> 7               | <input type="checkbox"/> FORD | <input type="checkbox"/> UNKNOWN | <input type="checkbox"/> REMOVED | Outlet Armoring | <input type="checkbox"/> NONE | <input type="checkbox"/> NOT EXTENSIVE | <input type="checkbox"/> EXTENSIVE |                               |                                        |                                   |                                |
|                       | Outlet Grade (Pick one)                                    | <input type="checkbox"/> AT STREAM GRADE <input type="checkbox"/> FREE FALL <input type="checkbox"/> CASCADE <input type="checkbox"/> FREE FALL ONTO CASCADE <input type="checkbox"/> CLOGGED/COLLAPSED/SUBMERGED <input type="checkbox"/> UNKNOWN            |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Outlet Dimensions                                          | A. Width                                                                                                                                                                                                                                                      |                            | B. Height                    |                            | C. Substrate/Water Width         |                              |                                          | D. Water Depth                |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Outlet Drop to Water Surface                               |                                                                                                                                                                                                                                                               |                            | Outlet Drop to Stream Bottom |                            |                                  |                              | E. Abutment Height (Type 7 bridges only) |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | L. Structure Length (Overall length from inlet to outlet)  |                                                                                                                                                                                                                                                               |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
| INLET                 | Inlet Shape                                                | <input type="checkbox"/> 1                                                                                                                                                                                                                                    | <input type="checkbox"/> 2 | <input type="checkbox"/> 3   | <input type="checkbox"/> 4 | <input type="checkbox"/> 5       | <input type="checkbox"/> 6   | <input type="checkbox"/> 7               | <input type="checkbox"/> FORD | <input type="checkbox"/> UNKNOWN | <input type="checkbox"/> REMOVED |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Inlet Type                                                 | <input type="checkbox"/> PROJECTING <input type="checkbox"/> HEADWALL <input type="checkbox"/> WINGWALLS <input type="checkbox"/> HEADWALL & WINGWALLS <input type="checkbox"/> MITERED TO SLOPE <input type="checkbox"/> OTHER <input type="checkbox"/> NONE |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Inlet Grade (Pick one)                                     | <input type="checkbox"/> AT STREAM GRADE <input type="checkbox"/> INLET DROP <input type="checkbox"/> PERCHED <input type="checkbox"/> CLOGGED/COLLAPSED/SUBMERGED <input type="checkbox"/> UNKNOWN                                                           |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Inlet Dimensions                                           | A. Width                                                                                                                                                                                                                                                      |                            | B. Height                    |                            | C. Substrate/Water Width         |                              |                                          | D. Water Depth                |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
| ADDITIONAL CONDITIONS | Slope % (Optional)                                         |                                                                                                                                                                                                                                                               |                            | Slope Confidence             |                            | <input type="checkbox"/> HIGH    | <input type="checkbox"/> LOW | Internal Structures                      |                               |                                  |                                  |                 |                               |                                        |                                    | <input type="checkbox"/> NONE | <input type="checkbox"/> BAFFLES/WEIRS | <input type="checkbox"/> SUPPORTS | <input type="checkbox"/> OTHER |
|                       | Structure Substrate Matches Stream                         | <input type="checkbox"/> NONE <input type="checkbox"/> COMPARABLE <input type="checkbox"/> CONTRASTING <input type="checkbox"/> NOT APPROPRIATE <input type="checkbox"/> UNKNOWN                                                                              |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Structure Substrate Type (Pick one)                        | <input type="checkbox"/> NONE <input type="checkbox"/> SILT <input type="checkbox"/> SAND <input type="checkbox"/> GRAVEL <input type="checkbox"/> COBBLE <input type="checkbox"/> BOULDER <input type="checkbox"/> BEDROCK <input type="checkbox"/> UNKNOWN  |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Structure Substrate Coverage                               | <input type="checkbox"/> NONE <input type="checkbox"/> 25% <input type="checkbox"/> 50% <input type="checkbox"/> 75% <input type="checkbox"/> 100% <input type="checkbox"/> UNKNOWN                                                                           |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Physical Barriers (Pick all that apply)                    | <input type="checkbox"/> NONE <input type="checkbox"/> DEBRIS/SEDIMENT/ROCK <input type="checkbox"/> DEFORMATION <input type="checkbox"/> FREE FALL <input type="checkbox"/> FENCING <input type="checkbox"/> DRY <input type="checkbox"/> OTHER              |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Severity (Choose carefully based on barrier type(s) above) | <input type="checkbox"/> NONE <input type="checkbox"/> MINOR <input type="checkbox"/> MODERATE <input type="checkbox"/> SEVERE                                                                                                                                |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Water Depth Matches Stream                                 | <input type="checkbox"/> YES <input type="checkbox"/> NO-SHALLOWER <input type="checkbox"/> NO-DEEPER <input type="checkbox"/> UNKNOWN <input type="checkbox"/> DRY                                                                                           |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Water Velocity Matches Stream                              | <input type="checkbox"/> YES <input type="checkbox"/> NO-FASTER <input type="checkbox"/> NO-SLOWER <input type="checkbox"/> UNKNOWN <input type="checkbox"/> DRY                                                                                              |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Dry Passage through Structure?                             | <input type="checkbox"/> YES                                                                                                                                                                                                                                  |                            | <input type="checkbox"/> NO  |                            | <input type="checkbox"/> UNKNOWN |                              | Height above Dry Passage                 |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |
|                       | Comments                                                   |                                                                                                                                                                                                                                                               |                            |                              |                            |                                  |                              |                                          |                               |                                  |                                  |                 |                               |                                        |                                    |                               |                                        |                                   |                                |

# Structure Shape & Dimensions

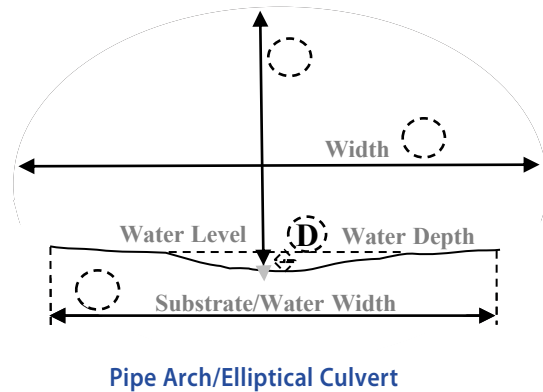
- 1) Select the Structure Shape number from the diagrams below and record it on the form for Inlet and Outlet Shape.
- 2) Record on the form in the appropriate blanks dimensions **A**, **B**, **C** and **D** as shown in the diagrams;  
**C** captures the width of water or substrate, whichever is wider; for dry culverts without substrate, C = 0.  
**D** is the depth of water -- be sure to measure inside the structure; for dry culverts, D = 0.
- 3) Record Structure Length (**L**). (Record abutment height (**E**) only for Type 7 Structures.)
- 4) For multiple culverts, also record the Inlet and Outlet shape and dimensions for each additional culvert.

**NOTE:** Culverts 1, 2 & 4 may or may not have substrate in them, so height measurements (B) are taken from the level of the "stream bed", whether that bed is composed of substrate or just the inside bottom surface of a culvert (grey arrows below show measuring to bottom, black arrows show measuring to substrate).

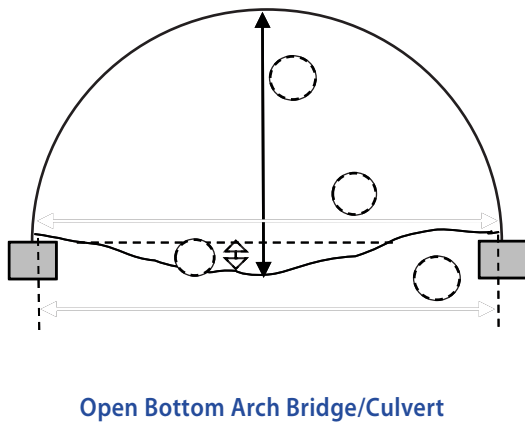
1



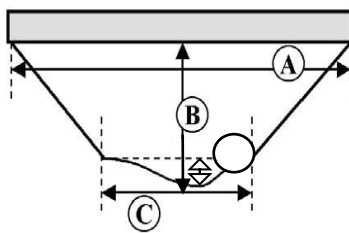
2



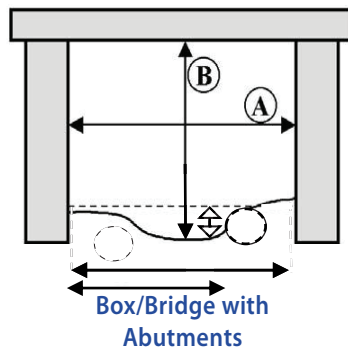
3



5



6



7

